

Times Series Part 1

Samuel Rowe adapted from Wooldridge (2011)

2025-12-02

Contents

Overview	5
1 Static Models	7
1.1 Static Philips Curve	7
1.2 Inflation and Deficits on Interest Rates	10
1.3 Puerto Rican Employment and Minimum Wage	12
1.4 Antidumping Filings and Chemical Imports	13
2 Finite Distributed Lags Models	15
2.1 Personal Exemption on Fertility Rates	15
3 Time Trends	19
3.1 Housing Investment and Housing Prices	19
3.2 Personal Exemption on Fertility Rates	21
3.3 Puerto Rican Employment	22
4 Seasonality	25
5 Weakly Dependent and Stationary Time Series AR(1) Models	27
5.1 Efficient Market Hypothesis	27
5.2 Expected Augmented Phillips Curve	29

bookdown::render_book()

Overview

Topics:

1. Static Model
2. Finite Distributed Lag Model
3. Time Trends
4. Seasonality
5. AR(1) Model

Chapter 1

Static Models

1.1 Static Philips Curve

Lesson: Problematic analysis with a static model between inflation and unemployment

Set the Time Series

```
cd "/Users/Sam/Desktop/Econ 645/Data/Wooldridge"
use phillips.dta, clear
tsset year, yearly
reg inf unem if year < 1997
```

/Users/Sam/Desktop/Econ 645/Data/Wooldridge

time variable: year, 1948 to 2003
delta: 1 year

Source	SS	df	MS	Number of obs	=	49
Model	25.6369575	1	25.6369575	F(1, 47)	=	2.62
Residual	460.61979	47	9.80042107	Prob > F	=	0.1125
Total	486.256748	48	10.1303489	R-squared	=	0.0527
				Adj R-squared	=	0.0326
				Root MSE	=	3.1306

inf	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
unem	.4676257	.2891262	1.62	0.112	-.1140213 1.049273
_cons	1.42361	1.719015	0.83	0.412	-2.034602 4.881822

Our result do not suggest a trade off between inflation and unemployment, and potentially suggest a positive relationship. This is likely a misspecified model and does not best describe the short-run trade-off between inflation and unemployment. We'll look at an augmented Phillips curve that better describes the relationship.

Let's graph our time series

```
twoway line inf year, title("Annual Inflation Rate")
graph export "/Users/Sam/Desktop/Econ 645/Stata/week_10_inf_line.png", replace
```

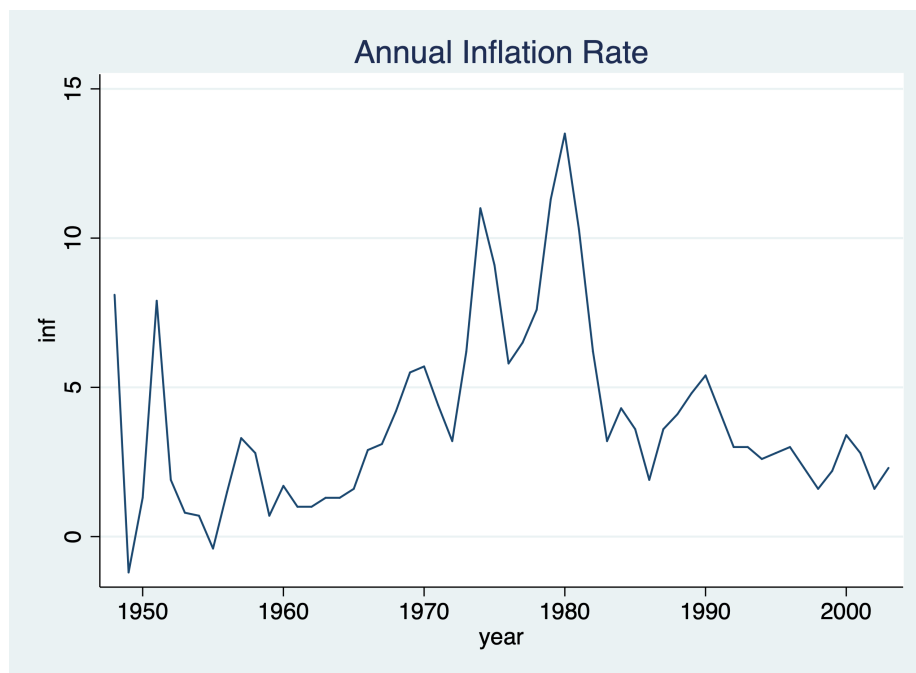


Figure 1.1: Line graph of inflation

We can use `tsline` as well.

```
tsline inf unem
graph export "/Users/Sam/Desktop/Econ 645/Stata/week_10_ts_infemp.png", replace
```

We can take the difference in the time series as well. Use the `s.` operator instead of `d.` While `s1` and `d1` will produce similar results, `s2` and `d2` will not produce the same results.

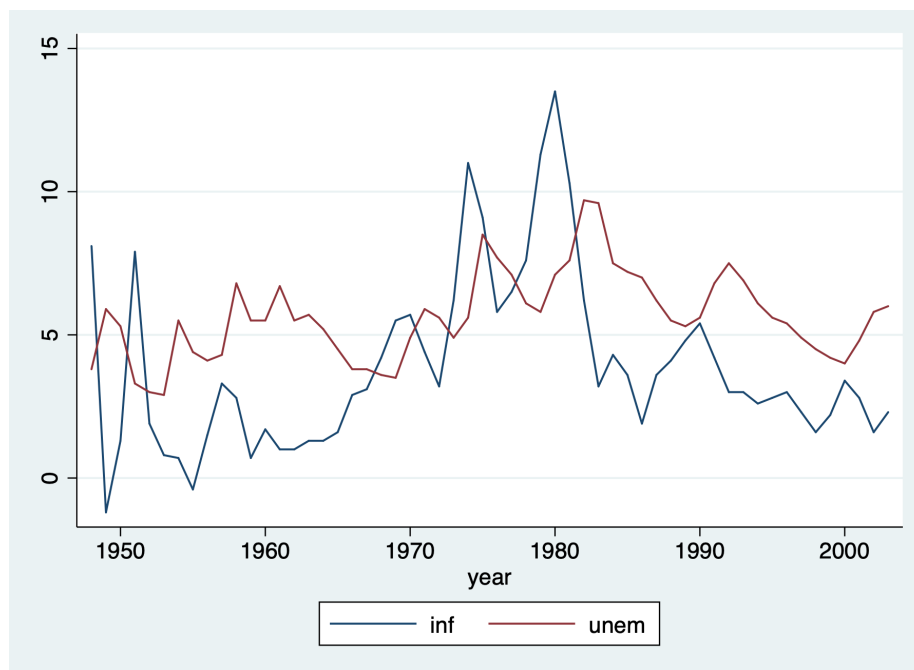


Figure 1.2: TSLINE graph of inflation and unemployment

```
tsline s.inf s.unem
graph export "/Users/Sam/Desktop/Econ 645/Stata/week_10_ts_Dinfemp.png", replace
```

Caution:

$$D2. \neq S2.$$

It is not recommended to use d. beyond the first difference. Use the S12. operator If you are interested in

$$x_t - x_{t-12} = \Delta x$$

$D2$ gives the difference in differences (not the research design. Such that the $D2$.operator calculates:

$$D2 = x_t - x_{t-1} - (x_{t-1} - x_{t-2})$$

$$S2 = x_t - x_{t-2}$$

We can take the 12-month difference in the time series as well.

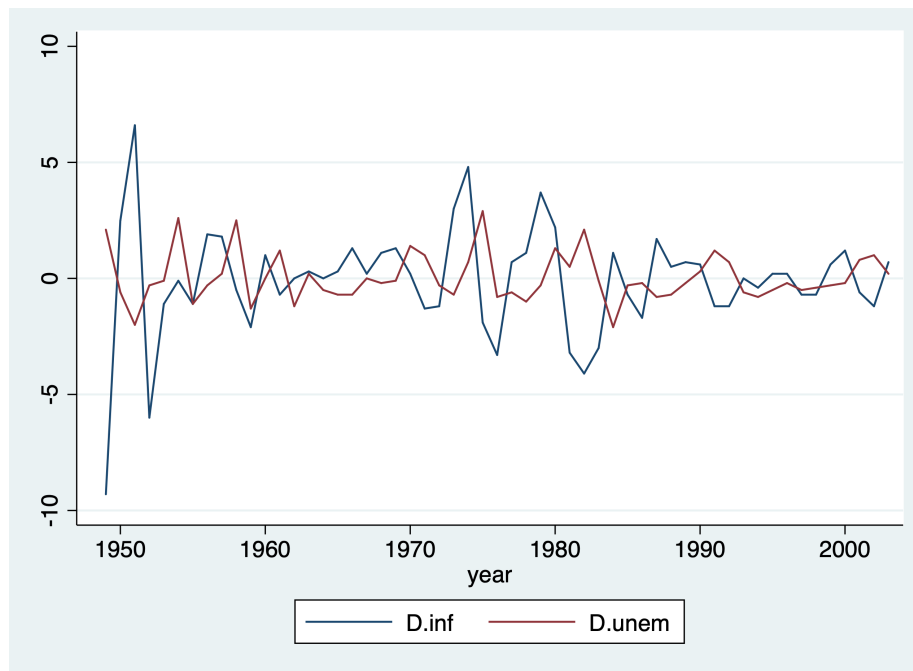


Figure 1.3: TSLINE graph of first difference in inflation and unemployment

```
tsline s2.inf s2.unem
graph export "/Users/Sam/Desktop/Econ 645/Stata/week_10_ts_D12infemp.png", replace
```

1.2 Inflation and Deficits on Interest Rates

Lesson: some static models explain time-series better than the prior model

Looking inflation and deficits on interest rates, our model is:

$$i3_t = \beta_0 + \beta_1 inflation_t + \beta_2 deficitpct_t + u_t$$

Where

1. $i3$: the 3-month T-bill rate
2. inf : the annual inflation rate from CPI
3. def : is the federal budget deficit as a percentage of GDP

Set the Time Series

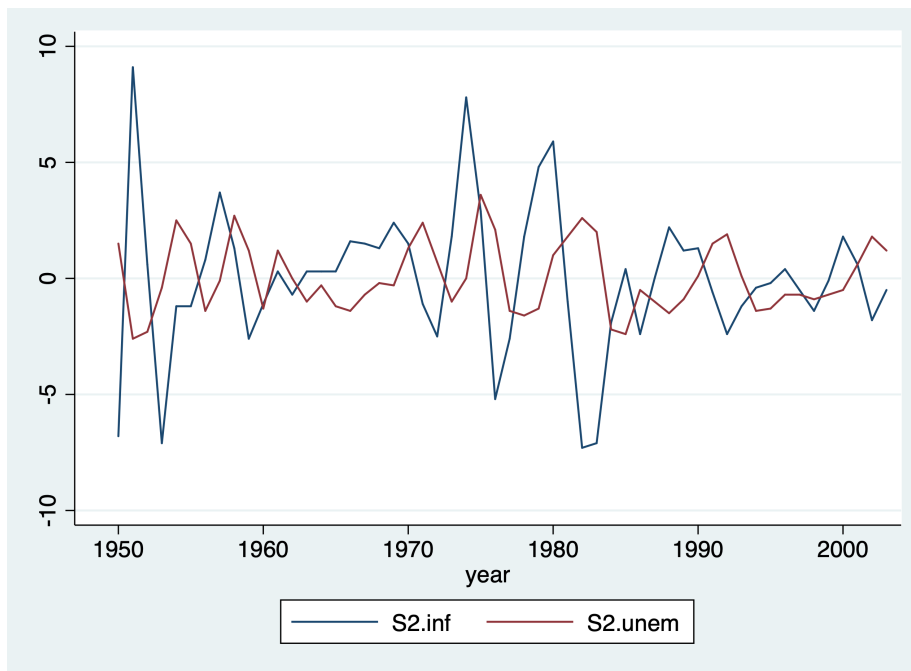


Figure 1.4: TSLINE graph of 2-year difference in inflation and unemployment

```
cd "/Users/Sam/Desktop/Econ 645/Data/Wooldridge"
use intdef.dta, clear
tsset year
reg i3 inf def if year < 2004
```

```
/Users/Sam/Desktop/Econ 645/Data/Wooldridge
```

```
time variable: year, 1948 to 2003
delta: 1 unit
```

Source	SS	df	MS	Number of obs	=	56
Model	272.420338	2	136.210169	F(2, 53)	=	40.09
Residual	180.054275	53	3.39725047	Prob > F	=	0.0000
				R-squared	=	0.6021
				Adj R-squared	=	0.5871
Total	452.474612	55	8.22681113	Root MSE	=	1.8432

i3	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]

inf	.6058659	.0821348	7.38	0.000	.4411243	.7706074
def	.5130579	.1183841	4.33	0.000	.2756095	.7505062
_cons	1.733266	.431967	4.01	0.000	.8668497	2.599682

This static model better explain interest rates based on basic macroeconomics. Both contemporaneous variables explain current interest rates. A 1 percentage point increase in the inflation rate leads to a 0.6 percentage point increase in the nominal interest rate. A 1 percentage point increase in the deficits as a percentage of GDP leads to a 0.5 percentage point increase in nominal interest rates.

1.3 Puerto Rican Employment and Minimum Wage

We'll look at the association between employment and minimum wage in Puerto Rico. Our model will be:

$$lprepop_t = \beta_0 + \beta_1 lmincov_t + \beta_2 lusgnp + u_t$$

Where

1. $\ln(\text{prepop})$: the natural log of the employment to population ratio in PR
2. $\ln(\text{mincov})$: natural log of the importance of minimum wage relative to average wages or $\text{mincov} = \text{average minimum wage} / \text{average wages}$ * average coverage rate or people actually covered by the minimum wage law
3. $\ln(\text{usgnp})$: natural log of US GNP

Set Time Series

```
cd "/Users/Sam/Desktop/Econ 645/Data/Wooldridge"
use prminwge, clear
tsset year
reg lprepop lmincov lusgnp if year < 1988
```

/Users/Sam/Desktop/Econ 645/Data/Wooldridge

```
time variable:  year, 1950 to 1987
delta: 1 unit
```

Source	SS	df	MS	Number of obs	=	38
-----+-----				F(2, 35)	=	34.04
Model	.211258366	2	.105629183	Prob > F	=	0.0000

Residual		.108600151	35	.003102861	R-squared	=	0.6605
-----+							
					Adj R-squared	=	0.6411
Total		.319858518	37	.008644825	Root MSE	=	.0557

lprepop		Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
-----+							
lmincov		-.1544442	.0649015	-2.38	0.023	-.2862011	-.0226872
lusgnp		-.0121888	.0885134	-0.14	0.891	-.1918806	.167503
_cons		-1.054423	.7654065	-1.38	0.177	-2.60828	.4994351

Our results show that there is a tradeoff between the employment ratio and the importance of minimum wage. A one percent increase in importance of minimum wage the employment-population ratio declines .154 percent.

1.4 Antidumping Filings and Chemical Imports

The barium industry in the U.S. complained to the International Trade Commission that China was dumping (or selling imports at lower prices to undercut domestic producers). First, are imports unusually high before the complaint? Second, do imports change after the dumping complaint?

Our model is the

$$lchnimp_t = \beta_0 + \beta_1 lchempi_t + \beta_2 lgas_t + \beta_3 lrtwex_t + \beta_4 befile6_t + \beta_5 affile6_t + \beta_6 afdec6_t + u_t$$

Where

1. lchnimp: natural log of the volume of Chinese imports of barium chloride
2. lchempi: natural log of domestic chemical production index
3. lgas: natural log of gasoline production
4. lrtwex: natural log of the exchange rate index
5. befile6: binary for 6 months before a complaint filing
6. affile6: binary for 6 months after a complaint filing
7. afdec6: binary for 6 months after a positive decision

Set Time Series Monthly

```
cd "/Users/Sam/Desktop/Econ 645/Data/Wooldridge"
use barium.dta, clear
tsset t, monthly
reg lchnimp lchempi lgas lrtwex befile6 affile6 afdec6
```

/Users/Sam/Desktop/Econ 645/Data/Wooldridge

time variable: t, 1960m2 to 1970m12
delta: 1 month

Source	SS	df	MS	Number of obs	=	131
Model	19.4051607	6	3.23419346	F(6, 124)	=	9.06
Residual	44.2470875	124	.356831351	Prob > F	=	0.0000
				R-squared	=	0.3049
				Adj R-squared	=	0.2712
Total	63.6522483	130	.489632679	Root MSE	=	.59735

lchnimp	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
lchempi	3.117193	.4792021	6.50	0.000	2.168718	4.065668
lgas	.1963504	.9066172	0.22	0.829	-1.598099	1.9908
lrtwex	.9830183	.4001537	2.46	0.015	.1910022	1.775034
befile6	.0595739	.2609699	0.23	0.820	-.4569585	.5761064
affile6	-.0324064	.2642973	-0.12	0.903	-.5555249	.490712
afdec6	-.565245	.2858352	-1.98	0.050	-1.130993	.0005028
_cons	-17.803	21.04537	-0.85	0.399	-59.45769	23.85169

We see that imports are not higher 6 months before a complaint, and imports are not lower 6 months after a complaint. However, imports are lower 6 months after a positive decision in a complaint by the ITC. The impact is fairly large and imports of barium chloride fall about 43.2% after a positive decision.

Also notice that rtwex is positive, and we would expect that a stronger dollar increase demand for imports, which is about uni-elastic.

```
display (exp(-.565)-1)*100
```

-43.163985

Chapter 2

Finite Distributed Lags Models

2.1 Personal Exemption on Fertility Rates

Lesson: using lags in the explanatory variables

Our interest the general fertility rate, which is the number of children born to every 1,000 women of childbearing age. PE is the average real dollar value of the personal tax exemption and two binary variables for World War II and the introduction of the birth control pill in 1963.

Our contemporaneous model is:

$$gfr_t = \beta_0 + \beta_1 pe_t + \beta_2 ww2_t + \beta_3 pill_t + u_t$$

Where

1. gfr: the fertility rate or births per 1,000 women at time t
2. pe: the personal tax exemptions at time t
3. ww2: a binary for World War II
4. pill: a binary for birth control pill

Set Time Series

```
cd "/Users/Sam/Desktop/Econ 645/Data/Wooldridge"  
use fertil3.dta, clear  
tsset year, yearly  
reg gfr pe i.ww2 i.pill if year < 1985
```

/Users/Sam/Desktop/Econ 645/Data/Wooldridge

time variable: year, 1913 to 1984
delta: 1 year

Source	SS	df	MS	Number of obs	=	72
Model	13183.6215	3	4394.54049	F(3, 68)	=	20.38
Residual	14664.2739	68	215.651087	Prob > F	=	0.0000
				R-squared	=	0.4734
				Adj R-squared	=	0.4502
Total	27847.8954	71	392.223879	Root MSE	=	14.685

gfr	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
pe	.08254	.0296462	2.78	0.007	.0233819	.1416981
1.ww2	-24.2384	7.458253	-3.25	0.002	-39.12111	-9.355684
1.pill	-31.59403	4.081068	-7.74	0.000	-39.73768	-23.45039
_cons	98.68176	3.208129	30.76	0.000	92.28003	105.0835

World War II reduced the fertility rate by 24.24 births per 1,000 women of child bearing age, while the introduction of the birth control pill reduced the number of birth by 31.59 births per 1,000 women of childbearing age. The personal tax exemption. A one dollar increase in the average real value of the personal tax exemption increases 0.083 births per 1,000 women of child bearing age or a 12 dollar increase in the average real personal tax exemption increases 1 child per 1,000 women of child bearing ago

Next lets add lags in the average real value of the personal tax exemption. Our Finite Distributed Lag model is:

$$gfr_t = \beta_0 + \beta_1 pe_t + \beta_2 pe_{t-1} + \beta_3 pe_{t-2} + \beta_4 ww2_t + \beta_5 pill_t + u_t$$

```
reg gfr pe pe_1 pe_2 i.ww2 i.pill if year < 1985
```

Source	SS	df	MS	Number of obs	=	70
Model	12959.7886	5	2591.95772	F(5, 64)	=	12.73
Residual	13032.6443	64	203.635067	Prob > F	=	0.0000
				R-squared	=	0.4986
				Adj R-squared	=	0.4594
Total	25992.4329	69	376.701926	Root MSE	=	14.27

gfr	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
-----	-------	-----------	---	------	----------------------	--

pe		.0726718	.1255331	0.58	0.565	-.1781094	.323453
pe_1		-.0057796	.1556629	-0.04	0.970	-.316752	.3051929
pe_2		.0338268	.1262574	0.27	0.790	-.2184013	.286055
1.ww2		-22.1265	10.73197	-2.06	0.043	-43.56608	-.6869196
1.pill		-31.30499	3.981559	-7.86	0.000	-39.25907	-23.35091
_cons		95.8705	3.281957	29.21	0.000	89.31403	102.427

Our personal exemption variables are no longer statistically significant, but let's test the joint significance. We might have multicollinearity that bias our standard errors for our PE variables.

```
test pe pe_1 pe_2
test pe_1 pe_2
```

```
( 1) pe = 0
( 2) pe_1 = 0
( 3) pe_2 = 0

F( 3, 64) = 3.97
Prob > F = 0.0117
```

```
( 1) pe_1 = 0
( 2) pe_2 = 0

F( 2, 64) = 0.05
Prob > F = 0.9480
```

Our PE variables are jointly significant, but our lags are jointly insignificant so we'll use our static model.

For the estimated LRP:

$$.073 - .0058 + 0.034 \approx .101,$$

but we lack a standard error. We'll need to estimate the standard error. Let

$$\theta_0 = \delta_0 + \delta_1 + \delta_2$$

denote the LRP and write

$$\delta_0$$

in terms of

$$\theta_0, \delta_1 \text{ and } \delta_2 \text{ where } \delta_0 = \theta_0 - \delta_1 - \delta_2$$

Next substitute for

$$\delta_0$$

$$gfr_t = \alpha_0 + \delta_0 pe_t + \delta_1 pe_{t-1} + \delta_2 pe_{t-2} + \dots$$

to get

$$gfr_t = \alpha_0 + (\theta_0 - \delta_1 - \delta_2) pe_t + \delta_1 pe_{t-1} + \delta_2 pe_{t-2} + \dots$$

$$gfr_t = \alpha_0 + \theta_0 pe_t + \delta_1 (pe_{t-1} - pe_t) + \delta_2 (pe_{t-2} - pe_t) +$$

We can estimate

$$\hat{\theta}_0$$

and its standard error. We can regress gfr_t onto pe_t, (pe_(t-1) - pe_t), and (pe_t-2 - pe_t), ww2, and pill

```
gen pe_1_1 = pe_1 - pe
gen pe_2_1 = pe_2 - pe
reg gfr pe pe_1_1 pe_2_1 i.ww2 i.pill
```

(1 missing value generated)

(2 missing values generated)

Source	SS	df	MS	Number of obs	=	70
				F(5, 64)	=	12.73
Model	12959.7886	5	2591.95772	Prob > F	=	0.0000
Residual	13032.6443	64	203.635067	R-squared	=	0.4986
				Adj R-squared	=	0.4594
Total	25992.4329	69	376.701926	Root MSE	=	14.27

gfr	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
pe	.1007191	.0298027	3.38	0.001	.0411814	.1602568
pe_1_1	-.0057796	.1556629	-0.04	0.970	-.316752	.3051929
pe_2_1	.0338268	.1262574	0.27	0.790	-.2184013	.286055
1.ww2	-22.1265	10.73197	-2.06	0.043	-43.56608	-.6869196
1.pill	-31.30499	3.981559	-7.86	0.000	-39.25907	-23.35091
_cons	95.8705	3.281957	29.21	0.000	89.31403	102.427

$\hat{\theta} \approx 0.101$ and significant, so our LRP has an effect on general fertility rates.

Chapter 3

Time Trends

3.1 Housing Investment and Housing Prices

Lesson: Importance of time trends

We look at annual housing investment and a housing price index from 1947 to 1988.

Set Time Series

```
cd "/Users/Sam/Desktop/Econ 645/Data/Wooldridge"
use hseinv.dta, clear
tsset year, yearly
```

```
/Users/Sam/Desktop/Econ 645/Data/Wooldridge
```

```
time variable: year, 1947 to 1988
delta: 1 year
```

Our model is the natural log of invpc or real per capita housing investment (\$1,000) and price be the housing price index where 1982 = 1 (or 100). We'll assume constant elasticity with our log-log model.

$$\ln invpc_t = \beta_0 + \beta_1 \ln price_t + u_t$$

```
reg lninvpc lnprice
```

Source	SS	df	MS	Number of obs	=	42
-----+-----				F(1, 40)	=	10.53

Model		.254364468	1	.254364468	Prob > F	=	0.0024
Residual		.966255566	40	.024156389	R-squared	=	0.2084
-----+					Adj R-squared	=	0.1886
Total		1.22062003	41	.02977122	Root MSE	=	.15542

linvpc		Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
-----+							
lprice		1.240943	.3824192	3.24	0.002	.4680452	2.013841
_cons		-.5502345	.0430266	-12.79	0.000	-.6371945	-.4632746

Our output shows that a 1 percent increase in the pricing index increases the investment per capita by 1.24 percent (or elastic response). Note: both investment and price have upward trends that we need to account for.

$$invpc_t = \beta_0 + \beta_1 lprice_t + \beta_2 t + u_t$$

```
reg linvpc lprice t
```

Source		SS	df	MS	Number of obs	=	42
-----+					F(2, 39)	=	10.08
Model		.415945108	2	.207972554	Prob > F	=	0.0003
Residual		.804674927	39	.02063269	R-squared	=	0.3408
-----+					Adj R-squared	=	0.3070
Total		1.22062003	41	.02977122	Root MSE	=	.14364

linvpc		Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
-----+							
lprice		-.3809612	.6788352	-0.56	0.578	-1.754035	.9921125
t		.0098287	.0035122	2.80	0.008	.0027246	.0169328
_cons		-.9130595	.1356133	-6.73	0.000	-1.187363	-.6387557

After accounting for the upward linear trend for both investment per capita and the housing price index, an increase in price is coefficient is now negative, but it is not statistically significant. We probably have serial correlation which will bias our standard errors.

```
predict u, resid
reg u l.u, noconst
drop u
```


gfr	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
pe	.2788778	.0400199	6.97	0.000	.1989978	.3587578
1.ww2	-35.59228	6.297377	-5.65	0.000	-48.1619	-23.02266
1.pill	.9974479	6.26163	0.16	0.874	-11.50082	13.49571
t	-1.149872	.1879038	-6.12	0.000	-1.524929	-.7748145
_cons	111.7694	3.357765	33.29	0.000	105.0673	118.4716

Quadratic Trend Fertility is declining but at a decreasing negatively rate (eventually a positive t^2 takes over t)

```
reg gfr pe i.ww2 i.pill c.t##c.t if year < 1985
```

Source	SS	df	MS	Number of obs	=	72
Model	20236.3981	5	4047.27961	F(5, 66)	=	35.09
Residual	7611.49734	66	115.325717	Prob > F	=	0.0000
Total	27847.8954	71	392.223879	R-squared	=	0.7267
				Adj R-squared	=	0.7060
				Root MSE	=	10.739

gfr	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
pe	.3478126	.0402599	8.64	0.000	.2674311	.428194
1.ww2	-35.88028	5.707921	-6.29	0.000	-47.27651	-24.48404
1.pill	-10.11972	6.336094	-1.60	0.115	-22.77014	2.530696
t	-2.531426	.3893863	-6.50	0.000	-3.308861	-1.753991
c.t#c.t	.0196126	.004971	3.95	0.000	.0096876	.0295377
_cons	124.0919	4.360738	28.46	0.000	115.3854	132.7984

The fertility rate exhibits upward and downward trends between 1913 and 1984. Interestingly, pe remains fairly robust after adding time trends. The linear time trend shows that general fertility rates are falling by 1.15 childs per 1,000 women of childbearing age for each additional year. However, the quadratic shows that the trend is negative but decreasing rate and will eventually become positive.

3.3 Puerto Rican Employment

The last linear trend is for employment-population ratio and the importance of minimum wage coverage. We'll add a time trend along with mincov and usgnp.

Set the Time Series

```
cd "/Users/Sam/Desktop/Econ 645/Data/Wooldridge"
use prminwge, clear
tsset year, yearly
```

```
/Users/Sam/Desktop/Econ 645/Data/Wooldridge
```

```
time variable: year, 1950 to 1987
delta: 1 year
```

Add linear trend

```
reg lprepop lmincov lusgnp t
```

Source	SS	df	MS	Number of obs	=	38
Model	.270948453	3	.090316151	F(3, 34)	=	62.78
Residual	.048910064	34	.001438531	Prob > F	=	0.0000
				R-squared	=	0.8471
				Adj R-squared	=	0.8336
Total	.319858518	37	.008644825	Root MSE	=	.03793

	lprepop	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
	lmincov	-.1686949	.0442463	-3.81	0.001	-.2586142 -.0787757
	lusgnp	1.057351	.1766369	5.99	0.000	.6983813 1.41632
	t	-.0323542	.0050227	-6.44	0.000	-.0425616 -.0221468
	_cons	-8.696298	1.295764	-6.71	0.000	-11.32961 -6.062988

The employment-population ratio does not have much of a downward or upward trend, but usgnp does. The linear trend of usgnp is about 3% per year, so an estimate of 1.06 in our model means that when usgnp increases by 1% above its long-term trend, employment-population ratio increases by about 1.06%.

Chapter 4

Seasonality

We'll add monthly variables to account for the fact that imports are not seasonally adjusted.

Set Time Series Monthly

```
cd "/Users/Sam/Desktop/Econ 645/Data/Wooldridge"
use barium.dta, clear
tsset t, monthly
reg lchnimp lchempi lgas lrtwex befile6 affile6 afdec6 ///
    feb mar apr may jun jul aug sep oct nov dec
```

/Users/Sam/Desktop/Econ 645/Data/Wooldridge

time variable: t, 1960m2 to 1970m12
delta: 1 month

Source	SS	df	MS	Number of obs	=	131
Model	22.8083523	17	1.34166778	F(17, 113)	=	3.71
Residual	40.843896	113	.361450407	Prob > F	=	0.0000
				R-squared	=	0.3583
				Adj R-squared	=	0.2618
Total	63.6522483	130	.489632679	Root MSE	=	.60121

lchnimp	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
lchempi	3.26506	.4929302	6.62	0.000	2.288476 4.241644
lgas	-1.278121	1.389008	-0.92	0.359	-4.029997 1.473755
lrtwex	.6630496	.4713038	1.41	0.162	-.2706882 1.596787

befile6		.1397024	.2668075	0.52	0.602	-.3888914	.6682962
affile6		.0126317	.2786866	0.05	0.964	-.5394967	.5647601
afdec6		-.5213006	.30195	-1.73	0.087	-1.119518	.0769168
feb		-.4177089	.3044445	-1.37	0.173	-1.020868	.1854505
mar		.0590528	.2647308	0.22	0.824	-.4654266	.5835322
apr		-.4514825	.2683865	-1.68	0.095	-.9832046	.0802396
may		.0333085	.2692425	0.12	0.902	-.5001093	.5667264
jun		-.2063321	.2692515	-0.77	0.445	-.7397679	.3271038
jul		.0038354	.2787665	0.01	0.989	-.5484513	.5561222
aug		-.1570652	.2779928	-0.56	0.573	-.7078191	.3936887
sep		-.1341606	.2676556	-0.50	0.617	-.6644348	.3961135
oct		.0516921	.2668512	0.19	0.847	-.4769883	.5803725
nov		-.24626	.2628272	-0.94	0.351	-.766968	.274448
dec		.1328368	.2714234	0.49	0.626	-.4049019	.6705755
_cons		16.77877	32.42865	0.52	0.606	-47.46824	81.02577

We compare all months to January as the base period, but let's test their joint significance.

```
test feb mar apr may jun jul aug sep oct nov dec
```

```
( 1) feb = 0
( 2) mar = 0
( 3) apr = 0
( 4) may = 0
( 5) jun = 0
( 6) jul = 0
( 7) aug = 0
( 8) sep = 0
( 9) oct = 0
(10) nov = 0
(11) dec = 0
```

```
F( 11, 113) = 0.86
Prob > F = 0.5852
```

We find that they are jointly insignificant and seasonality does not seem to be an issue with barium chloride imports.

Chapter 5

Weakly Dependent and Stationary Time Series AR(1) Models

5.1 Efficient Market Hypothesis

We'll test the a version of the efficient market hypothesis (EMH) by looking at weekly stock return from 1976 to 1989.

Set Time Series

```
cd "/Users/Sam/Desktop/Econ 645/Data/Wooldridge"  
use nyse, clear  
tsset t, weekly
```

```
/Users/Sam/Desktop/Econ 645/Data/Wooldridge
```

```
time variable:  t, 1960w2 to 1973w16  
delta: 1 week
```

Our dependent variable is the weekly percentage return on the New York Stock Exchange. A strict form of the EMH states that information observable to the market prior to week t should not help to predict the return during week t .

$$E(y_t | y_{t-1}, y_{t-2}, \dots) = E(y_t)$$

We will test EMH by specifying an AR(1) model and our hypothesis states that beta for y_{t-1} will be equal to 0. We'll assume that stock returns are serially uncorrelated, so we can safely assume that they are weakly dependent.

$$\text{return}_t = \beta_0 + \rho \text{return}_{t-1} + e_t$$

```
reg return l.return
```

```
*Or
```

```
reg return return_1
```

```
predict u, resid
```

```
reg u l.u, noconst
```

```
drop u
```

Source	SS	df	MS	Number of obs	=	689
-----+-----				F(1, 687)	=	2.40
Model	10.6866231	1	10.6866231	Prob > F	=	0.1218
Residual	3059.73817	687	4.45376735	R-squared	=	0.0035
-----+-----				Adj R-squared	=	0.0020
Total	3070.42479	688	4.46282673	Root MSE	=	2.1104

return	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
return						
L1.	.0588984	.0380231	1.55	0.122	-.0157569	.1335538
_cons	.179634	.0807419	2.22	0.026	.0211034	.3381646

Source	SS	df	MS	Number of obs	=	689
-----+-----				F(1, 687)	=	2.40
Model	10.6866231	1	10.6866231	Prob > F	=	0.1218
Residual	3059.73817	687	4.45376735	R-squared	=	0.0035
-----+-----				Adj R-squared	=	0.0020
Total	3070.42479	688	4.46282673	Root MSE	=	2.1104

return	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
return_1	.0588984	.0380231	1.55	0.122	-.0157569	.1335538
_cons	.179634	.0807419	2.22	0.026	.0211034	.3381646

(2 missing values generated)

Source	SS	df	MS	Number of obs	=	688
-----+-----				F(1, 687)	=	0.00

Model		.00603936	1	.00603936	Prob > F	=	0.9706
Residual		3059.08227	687	4.45281262	R-squared	=	0.0000
-----+							
Total		3059.08831	688	4.44634929	Adj R-squared	=	-0.0015
					Root MSE	=	2.1102

	u		Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
-----+							
	u						
	L1.		.001405	.0381496	0.04	0.971	-.0734987 .0763087

We cannot reject the null hypothesis under our model, but we do have some evidence of positive serial correlation.

5.2 Expected Augmented Phillips Curve

We'll revisit the Phillips curve

Set Time Series

```
cd "/Users/Sam/Desktop/Econ 645/Data/Wooldridge"
use phillips.dta, clear
tsset year
```

```
/Users/Sam/Desktop/Econ 645/Data/Wooldridge
```

```
time variable: year, 1948 to 2003
delta: 1 unit
```

A linear version of the expectations augmented Phillips curve can be written as

$$\ln f_t - \ln f_t^e = \beta_1(\text{unemp}_t - \mu_0) + e_t$$

Where μ_0 is the natural rate of unemployment and $\ln f^e$ is the expected rate of inflation formed in year $t-1$. The difference between actual unemployment and the natural rate is called cyclical unemployment, while the difference between inflation and expected inflation is called unanticipated inflation. Our e_t is called our supply shock. If there is a trade-off between inflation and unemployment our $\hat{\beta}_1$ will be negative.

An assumption is made about inflationary expectations and under adaptive expectations, the expected value of current inflation depends on recently observed inflation. The expected inflation will be assumed to be equal to last years inflation

$$\ln f_t - \ln f_{t-1} = \beta_0 + \beta_1 \text{unemp}_t + e_t$$

$$\Delta inf_t = \beta_0 + \beta_1 unemp_t + e_t$$

Where

$$\Delta inf_t = inf_t - inf_{t-1}$$

$$\beta_0 = -\beta_1 \mu_0$$

Since β_1 is expected to be negative and β_0 is expected to be positive.

Therefore, under adaptive expectations, the augmented Phillips curve relates the change in inflation to the level of unemployment and a supply shock e_t . We'll assume assumptions TSC.1-TSC.5 hold.

First Difference in the dependent variable or adaptive expectations of inflation.

```
reg cinf unem if year < 1997
```

```
*or
```

```
reg d.inf unem if year < 1997
```

Source	SS	df	MS	Number of obs	=	48
Model	33.3830007	1	33.3830007	F(1, 46)	=	5.56
Residual	276.305134	46	6.00663335	Prob > F	=	0.0227
				R-squared	=	0.1078
				Adj R-squared	=	0.0884
Total	309.688135	47	6.58910925	Root MSE	=	2.4508

cinf	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
unem	-.5425869	.2301559	-2.36	0.023	-1.005867	-.079307
_cons	3.030581	1.37681	2.20	0.033	.2592061	5.801955

Source	SS	df	MS	Number of obs	=	48
Model	33.3829996	1	33.3829996	F(1, 46)	=	5.56
Residual	276.305138	46	6.00663344	Prob > F	=	0.0227
				R-squared	=	0.1078
				Adj R-squared	=	0.0884
Total	309.688138	47	6.58910932	Root MSE	=	2.4508

D.inf	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
unem	-.5425869	.2301559	-2.36	0.023	-1.005867	-.079307
_cons	3.030581	1.37681	2.20	0.033	.259206	5.801955

The trade-off between unanticipated inflation and cyclical unemployment is seen in $\hat{\beta}_1$. A 1-point increase in unemployment decreases unanticipated inflation by about .54 points.

We can estimate the natural rate of unemployment by dividing $\hat{\beta}_0$ by negative of $\hat{\beta}_1$

$$\mu_0 = \hat{\beta}_0 / -\hat{\beta}_1$$

```
display 3.03/.5425
```

```
5.5852535
```

Plot the graph.

```
twoway line cinf year || line unem year, lpattern(dash) ///
    legend(order(1 "Difference in Inflation" 2 "Unemployment Rate")) ///
    title("Augmented Phillips Curve") ytitle(Percentage) xtitle(year)

graph export "week_10_augmented_phillips.png", replace
```

Mitchell, N.M. (2020). *Data Management Using Stata: A Practical Handbook*. (2nd ed.). Stata Press.

UCLA Advanced Research Computing. (2025). *How can I access information stored after I run a command in Stata*. Retrieved from <https://stats.oarc.ucla.edu/stata/faq/how-can-i-access-information-stored-after-i-run-a-command-in-stata-returned-results/>

Wooldridge (2009). *Introductory Econometrics: A Modern Approach*. (4th ed.). South-Western Cengage Learning.

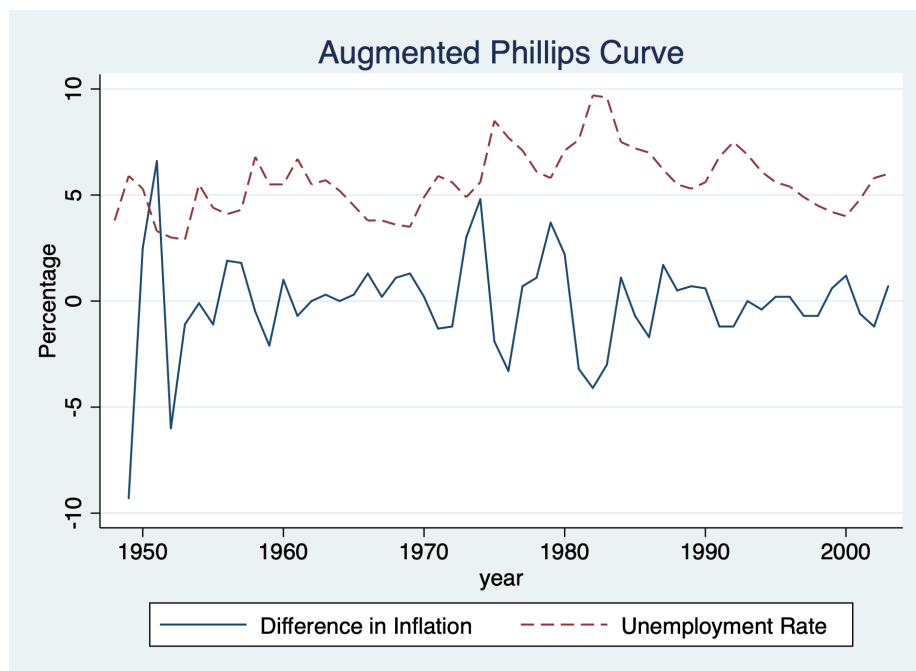


Figure 5.1: Line graph of inflation